# Quantum Mechanics Homework 7

Lectures 15–16

**Instructions.** Show the main steps of every calculation and briefly explain the physical meaning of your results. There are three problems, each worth 20 points. You may use the formulas below without deriving them.

## Useful formulas

For $b > 0$,

$$\int_{-\infty}^{\infty} e^{-bx^2}\,dx = \sqrt{\frac{\pi}{b}}, \tag{1}$$

$$\int_{-\infty}^{\infty} x e^{-bx^2}\,dx = 0, \tag{2}$$

$$\int_{-\infty}^{\infty} x^2 e^{-bx^2}\,dx = \frac{\sqrt{\pi}}{2b^{3/2}}. \tag{3}$$

For a normalized wave function,

$$\langle \hat{O} \rangle = \int_{-\infty}^{\infty} \psi^*(x)\hat{O}\psi(x)\,dx, \qquad (\Delta O)^2 = \langle \hat{O}^2 \rangle - \langle \hat{O} \rangle^2. \tag{4}$$

The momentum operator, free-particle Hamiltonian, and probability current are

$$\hat{p} = -i\hbar\frac{d}{dx}, \qquad \hat{H} = \frac{\hat{p}^2}{2m}, \qquad j = \frac{\hbar}{2mi}\left(\psi^*\frac{\partial\psi}{\partial x} - \psi\frac{\partial\psi^*}{\partial x}\right). \tag{5}$$

## Problem 1. A moving Gaussian wave packet [20 points]

At $t = 0$, a normalized free-particle state is

$$\psi(x,0) = \frac{1}{(\pi a^2)^{1/4}} \exp\left[-\frac{(x-x_0)^2}{2a^2}\right] e^{ik_0 x}, \qquad a > 0. \tag{6}$$

Define $p_0 = \hbar k_0$ and

$$E_0 = \frac{p_0^2}{2m} = \frac{\hbar^2 k_0^2}{2m}. \tag{7}$$

(a) Verify the normalization of $\psi(x,0)$. Calculate $\langle x \rangle$, $\langle x^2 \rangle$, and $\Delta x$. (5 points)

(b) Show that

$$(\hat{p} - p_0)\psi = i\hbar\frac{x - x_0}{a^2}\psi.$$

Use this result to calculate $\langle p \rangle$ and $\Delta p$. Verify that the packet has the minimum uncertainty product. (6 points)

(c) Calculate the mean free-particle energy $\langle H \rangle$. Clearly distinguish $E_0$ from $\langle H \rangle$, and explain why the latter is larger. (5 points)

(d) The free-particle dispersion relation is

$$\omega(k) = \frac{\hbar k^2}{2m}.$$

Find the group velocity $v_g$ and phase velocity $v_p$ at $k_0$. Write the packet-center trajectory $x_c(t)$ and the dispersion time $\tau$. (4 points)

## Problem 2. A potential step below the step height [20 points]

Consider

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & x > 0, \end{cases} \qquad 0 < E < V_0. \tag{8}$$

Define

$$k = \frac{\sqrt{2mE}}{\hbar}, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}. \tag{9}$$

A stationary wave is incident from the left.

(a) Write the general stationary solutions in the two regions. Explain why both $e^{ikx}$ and $e^{-ikx}$ must be kept for $x < 0$, but the growing exponential must be removed for $x > 0$. Hence write the physical scattering solution in the form

$$\phi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0, \\ De^{-\kappa x}, & x > 0. \end{cases}$$

(5 points)

(b) Apply continuity of $\phi$ and $\phi'$ at $x = 0$. Define

$$r = \frac{B}{A}, \qquad t = \frac{D}{A},$$

and derive

$$r = \frac{k - i\kappa}{k + i\kappa}, \qquad t = \frac{2k}{k + i\kappa}.$$

(6 points)

(c) Now take $E = V_0/2$. Find $k/\kappa$, $r$, and $t$. (4 points)

(d) Calculate the incident, reflected, and right-region currents. Find $R$ and $T$, and determine the penetration length $\ell = 1/\kappa$ for $E = V_0/2$. Explain why $t \neq 0$ does not imply transmission to $+\infty$. (5 points)

## Problem 3. A potential step above the step height [20 points]

For the same potential step, let $E > V_0$, and define

$$k_1 = \frac{\sqrt{2mE}}{\hbar}, \qquad k_2 = \frac{\sqrt{2m(E - V_0)}}{\hbar}. \tag{10}$$

There is no incoming wave from $+\infty$, so use

$$\phi(x) = \begin{cases} Ae^{ik_1 x} + Be^{-ik_1 x}, & x < 0, \\ Ce^{ik_2 x}, & x > 0. \end{cases} \tag{11}$$

(a) Apply continuity of $\phi$ and $\phi'$ at $x = 0$. With

$$r = \frac{B}{A}, \qquad t = \frac{C}{A},$$

derive

$$r = \frac{k_1 - k_2}{k_1 + k_2}, \qquad t = \frac{2k_1}{k_1 + k_2}.$$

(6 points)

(b) Use probability currents to show that

$$R = |r|^2, \qquad T = \frac{k_2}{k_1}|t|^2.$$

Verify directly that $R + T = 1$. (6 points)

(c) Take $E = 4V_0/3$. Find $k_2/k_1$, $r$, $t$, $R$, and $T$. Explain why $|t|^2$ is not the transmission probability in this case. (5 points)

(d) For the energy in part (c), compare the transmitted and incident wavelengths and group velocities. Why is there still some reflection even though $E > V_0$? (3 points)